k-Combinations

- **Definition**
  A ***k*-combination** of a set of n distinct elements is an unordered selection of $k \leq n$ elements from the set.

- **Example 10**
  Let $S = \{1, 2, 3, 4\}$. Then $\{1, 3, 4\}$ is a 3-combination from S. Moreover, $\{1, 4, 3\}, \{4, 3, 1\}\{4, 1, 3\}, \{3, 1, 4\}, \{3, 4, 1\}$ all refer to the same combination (they count as one possible solution).